CASE STUDY

Triosephosphate isomerase: the first proton transfer

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run with the 3.2.0 development version.

Triosephosphate isomerase (TIM) interconverts dihydroxyacetone phosphate (DHAP) and D-glyceraldehyde 3-phosphate (GAP), close to the diffusion limit. In the first step the carboxylate of **Glu165** removes a proton from C1 of DHAP and an enediolate intermediate is formed, stabilized by **His95** (a hydrogen bond to the carbonyl oxygen) and by Lys12:

DHAP + Glu165–COO^- → enediolate + Glu165–COOH

This case study scans that proton transfer with a QC/MM potential.

The system

The prepared system is `examples/tutorials/case_studies/1_tim/TIM_DHAP_AMBER_AM1.pkl` in the EasyHybrid folder. It was built from PDB **7TIM** (yeast TIM with the intermediate analogue phosphoglycolohydroxamate):

Model	chain A; the analogue rebuilt as DHAP
Force fields	AMBER ff99SB (protein), TIP3P (water), GAFF (DHAP)
Atoms	4975 (the solvated enzyme pruned around the active site)
Fixed atoms	3385; 1590 free atoms around the substrate
QC region	DHAP and the side chains of His95 and Glu165: 47 atoms, **AM1**, charge −3

(In the file, His95 and Glu165 are numbered HIE 94 and GLU 164.) Open it with **File → Open**; the status bar shows `hamiltonian: am1`, `QC atoms: 47`, `QC charge: -3`, `fixed atoms: 3385`.

The reaction coordinate

The proton H02 moves from C02 (C1 of DHAP) to OE2 of Glu165. As in the SN2 tutorial, the coordinate is a **multiple distance** with the transferred atom in the middle:

RC = d(C02–H02) − d(H02–OE2)

Pick, in **Picking** mode, **C02** (#1, atom 3337), **H02** (#2, atom 3338) and **OE2** of Glu165 (#3, atom 2219). In the reactant, C–H = 1.13 Å and H···O = 2.14 Å.

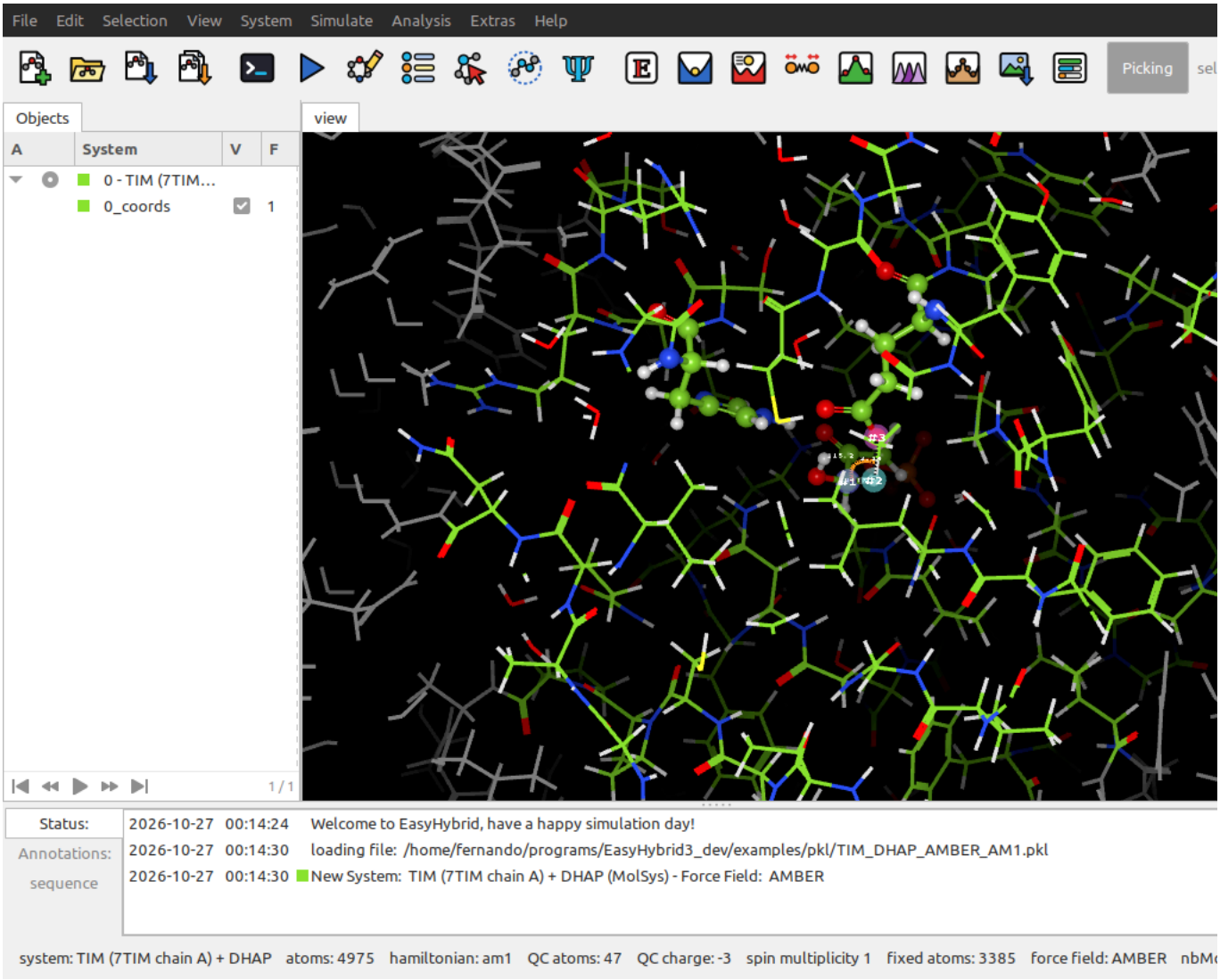


Figure 1. The active site: DHAP, His95 and Glu165 (balls and sticks, the QC region) with C02, H02 and OE2 picked.

The scan

Simulate → Reaction Coordinate Scans, **Coordinate Type** *multiple distance*, **Import from Picking Selection**:

Field	Value
Apply Mass Weighted Restraints	on (C and O at the ends: weights 0.43 and 0.57)
Step Size	0.1 Å
Number of Steps	16
Force Constant	4000 kJ mol^{-1} $Å^{-2}$
Method / Maximum iterations / RMS Grad. Tolerance	Conjugate Gradient / 600 / 0.1

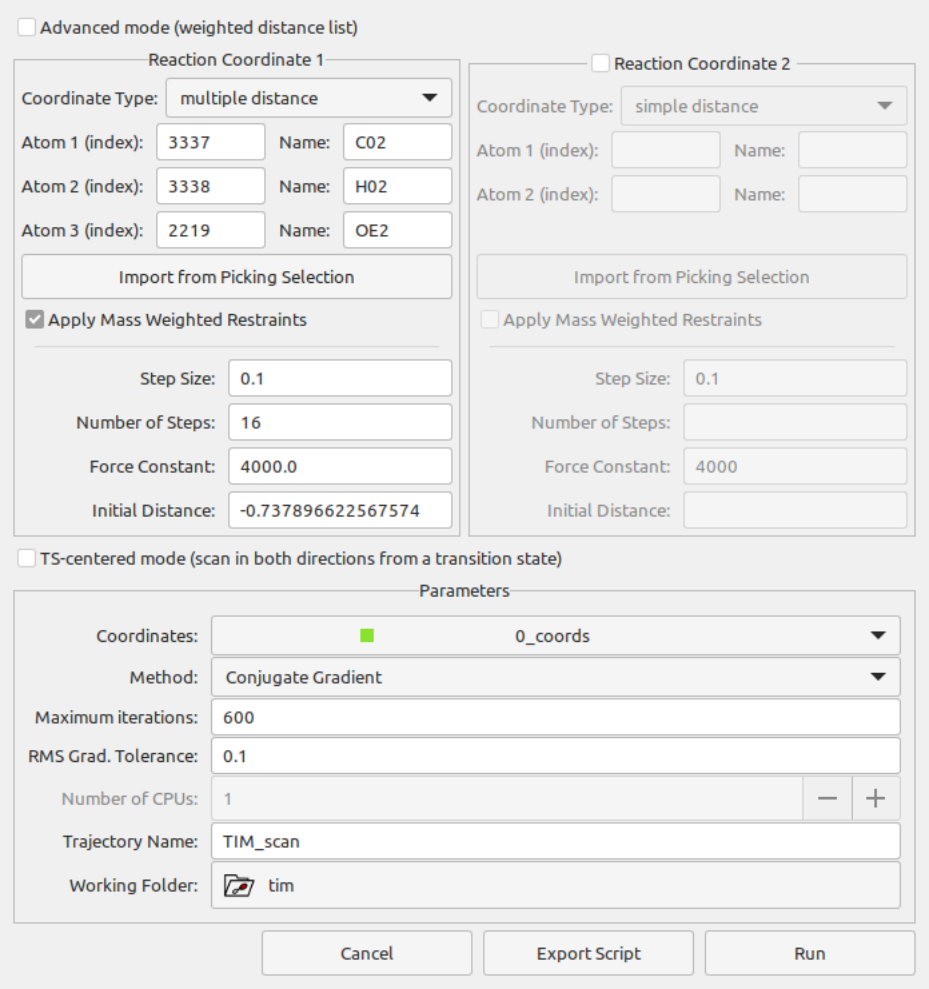


Figure 2. The proton transfer coordinate C02–H02–OE2.

Each point optimizes the 1590 free atoms with the QC/MM potential: the scan takes about 4 minutes on one core. Import it from the *Finished!* dialog and open **Analysis → PES Analysis**.

Results

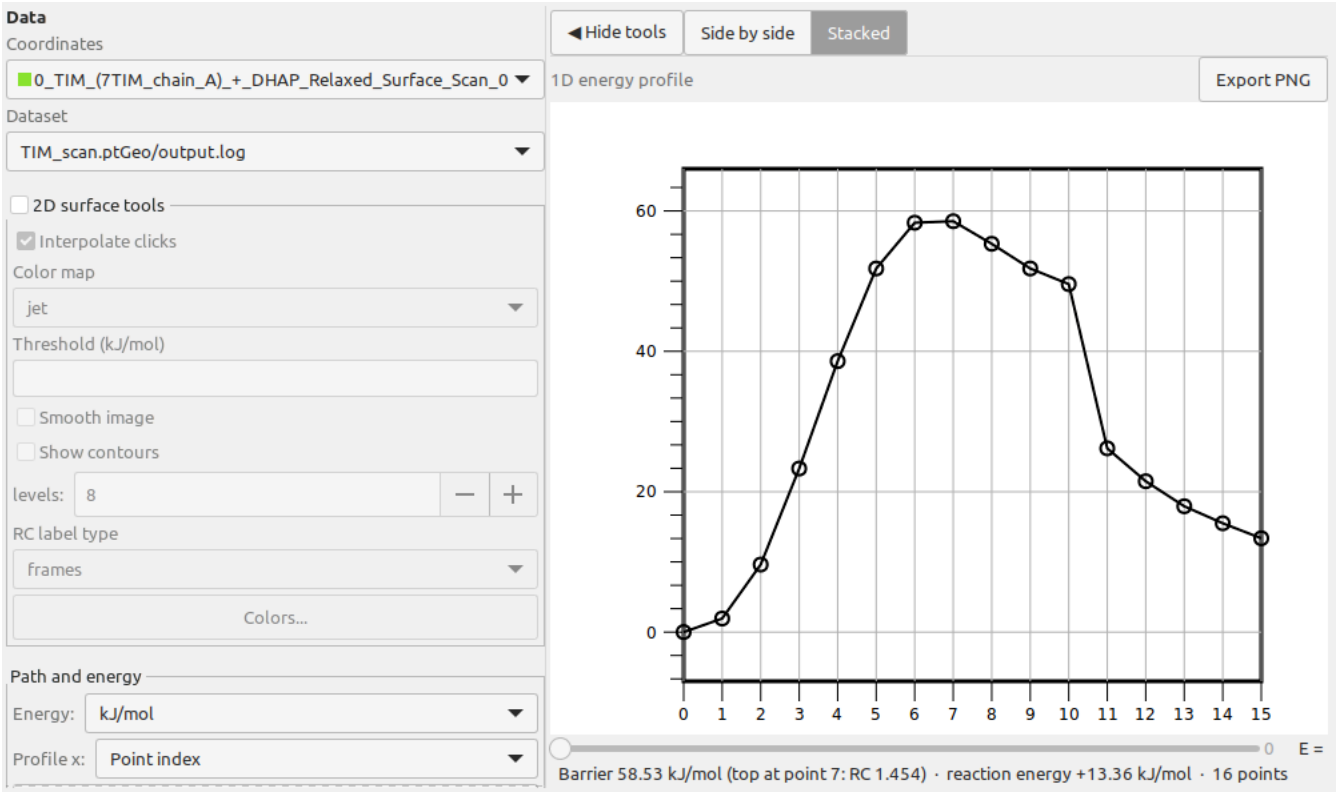


Figure 3. AM1/AMBER energy along the proton transfer (relative to the reactant).

Point	C02–H02 (Å)	H02–OE2 (Å)	E (kJ/mol)	
0	1.13	2.14	0.0	DHAP, Glu165⁻
7	1.45	1.15	**58.5**	proton in flight (highest point)
10	1.95	1.00	49.6	proton on Glu165
11	2.16	0.99	26.2	same, after the water rearranged
15	3.09	0.99	13.4	enediolate, Glu165–COOH

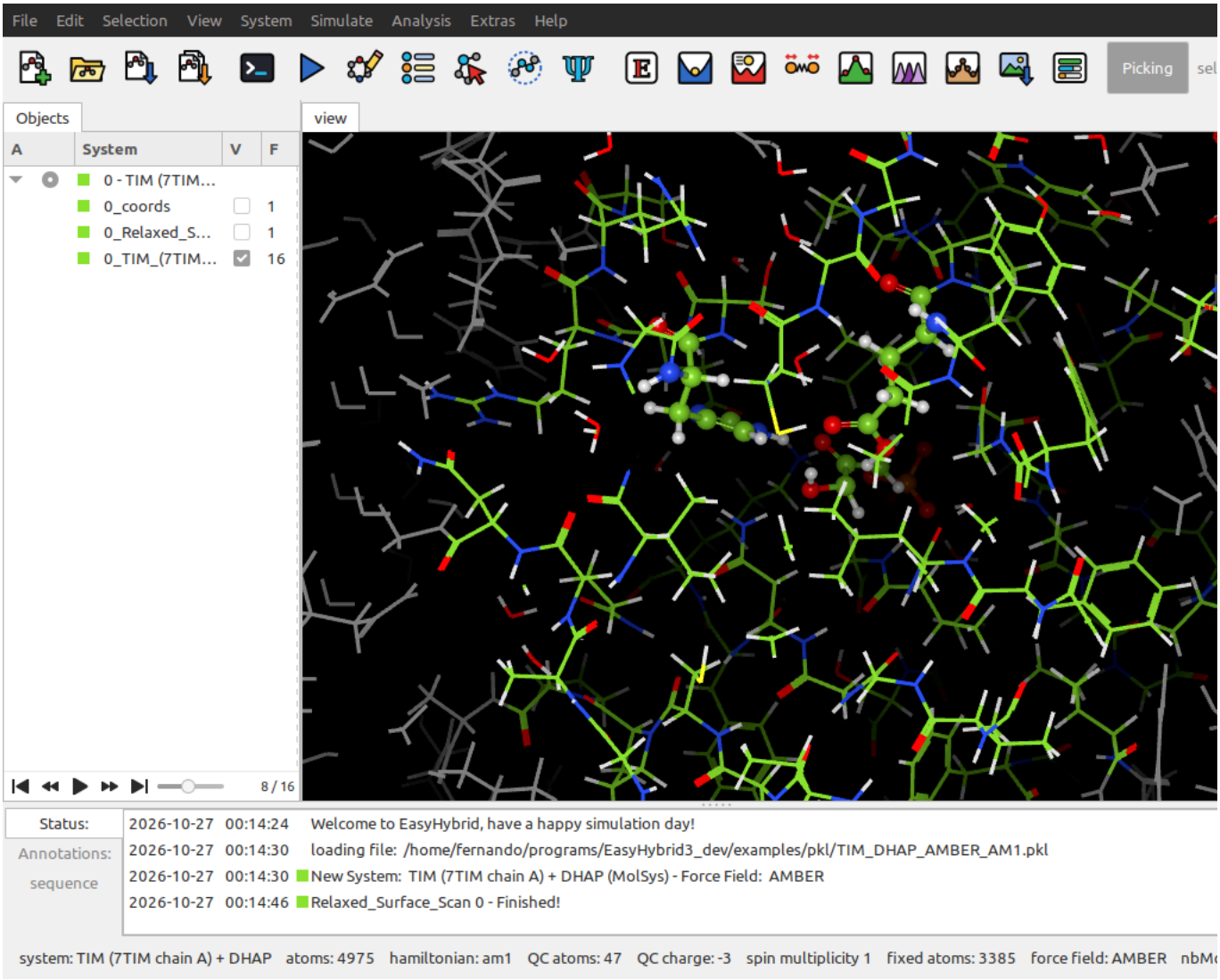


Figure 4. Point 7, the highest point: the proton between C1 of DHAP and Glu165.

The highest point, 58.5 kJ/mol (14 kcal/mol), has the proton almost halfway, slightly closer to the oxygen. The enediolate with protonated Glu165 lies about 13 kJ/mol above the reactant. For comparison, the k_cat of yeast TIM (about 4000 s^{-1}) corresponds to an overall free energy barrier of about 13 kcal/mol (54 kJ/mol). The numbers here are AM1 potential energies of single relaxed structures: a semiempirical method and no entropy, a first picture of the step.

The drop between points 10 and 11 (−23 kJ/mol) is not chemistry: the structures differ by the position of a few water molecules 13–20 Å from the proton, which moved to a lower minimum during the optimization of point 11. Relaxed scans of large systems often show such jumps: the environment falls into different local minima along the path. Their signatures are a jump in the energy with no change in the reacting atoms; the remedies are to scan back from the end and compare, to optimize more tightly, or to compute free energies (umbrella sampling) instead of single structures.

Notes

- **Size of the QC region.** Whole residues in the QC region (as His95 and Glu165 here) cut the backbone at polar bonds; a smaller region with the side chains cut at CA–CB is the usual choice (QC/MM tutorial).
- **The second step** (the enediolate gives the proton back to C2 through His95 and Glu165) can be scanned the same way from the last structure of this scan.
- References: J. R. Knowles, *Nature* **1991**, 350, 121 (the TIM mechanism); E. A. Komives, L. C. Chang, E. Lolis, R. F. Tilton, G. A. Petsko, J. R. Knowles, *Biochemistry* **1991**, 30, 3011 (the role of His95).

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3